Zachary Leong

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Education

University of Pennsylvania Philadelphia, PA

B.S.E. in Computer Science, major in Digital Media Design *May 2028*

- Coursework: Data Structures and Algorithms, Operating Systems, Computer Systems, Linear Algebra, 3D Modeling, Computer Animation, Procedural Graphics

M.S.E. in Computer Graphics & Game Technology *May 2028*

- Coursework: GPU Programming and Architecture, Offline & Real-time Rendering

Experience

Aviation Systems Software Engineering Intern June 2026 – August 2026

NASA Ames Research Center *Mountain View, CA*

- Architected a Python modernization of Freddie for drone traffic management, designing a distributed back-end of 4 microservices with custom Kafka, REST, WebSocket, and Postgres integrations for real-time data flow
- Built telemetry simulation, conformance checking, and visualizations for flight paths and 4D airspace volumes
- Developed a custom Python API bridging Freddie with a node-based simulation platform (ATM TestBed), generating 4D spatial constraint volumes from historical flight data to evaluate proposed drone trajectories

Teaching Assistant January 2026 – Present

Interactive Computer Graphics *Philadelphia, PA*

- Debug C++/OpenGL projects with students: scene graph, rasterization, GLSL shaders, and half-edge meshes
- Teach core concepts to 80+ students and review C++ implementations for design, quality, and performance

Research Intern October 2024 – December 2025

GRASP Sung Robotics Lab *Philadelphia, PA*

- Independent Study: *GPU-Accelerated 3D CSC Dubins Path Distance Checking using Signed Distance Fields*
  - Benchmarked scaling across 50+ links and 10M+ points, validating GPU scalability over CPU methods
- Implemented custom 3D transform gizmos for precise local/world-space manipulation of kinematic linkages using PyQtGraph, supporting a research paper submitted to *IROS 2026*

Projects

Fractal Path Tracer | *C++, CUDA, Nsight* Github

- Engineered a CUDA wavefront path tracer with Disney BSDF, fractal ray marching, and material sorting
- Optimized multiple importance sampling and next-event estimation using a separate shadow ray kernel
- Profiled extensively using Nsight Compute and used CUDA intrinsics in intersection hotpaths for 4x speedup

Katara Engine | *Rust, WebGPU, WGSL, Nsight* Live Demo · GitHub

- Built a real-time Smoothed-Particle Hydrodynamics (SPH) fluid engine for the web using Rust and wgpu
- Migrated pipeline to compute shaders, eliminating CPU-GPU readbacks to achieve 30 FPS on a laptop 4060
- Scaled to 600k+ particles via spatial hashing and particle sorting to improve memory coalescing

Space Minecraft | *C++, GLSL, OpenGL, Qt* Video Demo · Project

- Collaborated in a team of three to create a voxel game engine supporting spherical planets in a solar system
- Contributed debug GUI, shadow maps, biome blending, procedural planets, and foliage scattering
- Optimized performance by using compute shaders, multithreading, terrain chunking, and console profiling

Additional Links: Spring 2026 Demo Reel, Complex Origami Showcase

Technical Skills

Languages: C++, Rust, Python, Java, HTML, CSS, JavaScript, C, C#, OCaml, VEX, HLSL, GLSL, WGSL, Slang
Platforms/Tools: Vulkan, OpenGL, WebGPU, CUDA, Nsight Graphics/Compute, RenderDoc, Linux, Git, Claude Code
Software: Houdini, Blender, Maya, Substance Painter, DaVinci Resolve, Photoshop, Figma, Unity, Unreal Engine